PES UNIVERSITY, BANGALORE
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UE21CS341A - Software Engineering

Project Plan Document

*"**Happy Journey**: Your All-in-One Travel Companion App 🌍✈️🍽️ – Explore, Plan, and Connect with Fellow Travelers!"*

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LIFECYCLE FOLLOWED FOR HAPPY JOURNEY!:

AGILE

Justification for Choosing the Agile Lifecycle Model:

- **Adaptability to Changing Requirements:**
  Agile is well-suited for projects like "Happy Journey" where requirements may evolve or change during the development process. In the travel industry, market trends, user preferences, and external factors can influence feature priorities. Agile allows for flexibility in accommodating these changes without causing major disruptions to the project.

- **Frequent User Feedback**:
  Agile emphasizes continuous user involvement and feedback throughout the project. In the case of a travel companion app, user experience is paramount. Agile's iterative approach ensures that users' needs and preferences are integrated into the app's development, resulting in a product that better aligns with their expectations.

- **Incremental Delivery:**
  Agile promotes incremental development, meaning that working, tested increments of the product are delivered in short cycles (sprints). This approach allows for early releases of essential features, which can be particularly beneficial for a travel app. Users can start benefiting from the app sooner, and the development team can gather user feedback quickly.

- **Effective Risk Management:**

  Agile encourages risk mitigation by addressing potential issues early in the project. In the travel industry, factors like changing travel regulations or unforeseen disruptions (e.g., weather events) can impact project timelines. Agile's iterative nature allows for the identification and mitigation of risks at an early stage.

- **Collaborative Teamwork**:
  Agile promotes collaboration among cross-functional teams, including developers, designers, testers, and product owners. In a travel app project, this collaborative approach ensures that various aspects, such as user experience design, functionality, and quality, are all given due consideration.

- **Continuous Improvement**:
  Agile encourages retrospectives at the end of each sprint, where the team reflects on what went well and what can be improved. This iterative feedback loop leads to continuous improvement in both the product and the development process, making it well-suited for enhancing the "Happy Journey" app over time.

- **Enhanced User Engagement**:

Agile allows for close interaction between the development team and users, fostering a sense of ownership and engagement. In the context of a travel app, engaged users can become advocates, contributing to the app's growth and success.

TOOLS USED FOR HAPPY JOURNEY!:

Planning Tool:

- GitHub (Used for project management, collaboration among team members)
- JIRA(for managing, task tracking and time to time updates for all the team members)

Design Tool:

- Figma (For creating wireframes and prototypes for the user interface)
- Sketch, or Adobe XD for user interface and user experience design.

Version Control:

- Git and GitHub for source code version control (To track changes in the codebase and facilitate collaboration)

Development Tool:

- React JS (Frontend development along with HTML and CSS)
- MongoDB and also FireBase (Database management)
- NodeJS and ExpressJS (Backend development)
- VS Code (Preferred code editor for development)

Bug Tracking:

- JIRA(For management and tracking)
- GitHub Issues for tracking and managing software issues.

Testing Tool:

- JIRA(Testcase Management and Issue tracking)
- Selenium

DELIVERABLES USED FOR HAPPY JOURNEY!:

Reusable Components:

- **Libraries:**

  Justification:
  Libraries are software modules or packages that can be used in future projects. For example, utility libraries for handling common tasks like data storage or user authentication can be reused to save development time and maintain consistency across projects.

- **APIs (Application Programming Interfaces):**

  Justification:
  APIs allow other software systems to interact with specific functionalities of the "Happy Journey" . By making these APIs available, they can be reused to integrate the website's features into other applications, expanding its reach and usability.

- **Design Assets:**

  Justification:
  Design assets include UI/UX designs, templates, and style guides that serve as visual foundations for future app designs. Reusing these assets can ensure a consistent and appealing user experience in future projects, reducing design efforts.

- **Database assets:**

  Justification:
  If Happy Journey has the database for the attractions or restaurants then that can be reused for the future projects as this is already cleaned database for the HAPPY_JOURNEY!

Build Components (Specific Features and Functionalities of "Happy Journey"):

- **User Profiles:**

  Justification:
  User profiles are a core component of the app and provide personalization for users. They are specific to the "Happy Journey" app and are essential for delivering a customized travel experience.

- **Restaurant Discovery:**

  Justification:
  The restaurant discovery feature is a unique and integral part of the app, allowing users to explore local cuisine. It is specific to the "Happy Journey" app and adds

significant value to the user experience.

- **Attractions:**

  Justification:
  The Attraction feature is app-specific and enhances the unique travel experience provided by "Happy Journey," it adds substantial value to the app's users by offering them a comprehensive guide to local attractions. This feature enriches the travel experience, encourages user engagement, and reinforces the app's goal of becoming an all-in-one travel companion.

- **Real-Time Weather Updates:**

  Justification:
  This feature is specific to the app's functionality, it serves as a critical utility that enhances user travel planning and decision-making. Accurate weather data ensures that users can make informed choices regarding their travel activities, packing, and itinerary adjustments. This feature aligns with the app's mission to offer a holistic and convenient travel experience, making it a valuable and essential component.

- **Community Interaction:**

  Justification:
  Community interaction features are tailored to the "Happy Journey" app's goal of connecting travelers. They foster engagement and sharing within the app's user community, making them app-specific.

- **Travel Booking:**

  Justification:
  Travel booking features enable users to book flights, hotels, and travel packages directly through the app. While they are app-specific, they provide essential functionality and revenue generation for the project.

- **Search for Travel Partners:**

  Justification:
  Searching for travel partners is a unique feature designed to enhance the user experience within the "Happy Journey" app. It facilitates connections between users with similar travel interests and is not typically reused in other projects.

Project: "Happy Journey" - Work Breakdown Structure (WBS)

1. Project Initiation and Planning:
    - Define Project Scope and Objectives
    - Assemble Project Team
    - Develop Project Charter
    - Identify Risks and Mitigation Strategies
    - Create Project Timeline and Milestones
2. Requirements Gathering and Analysis:
    - Perform Market Research
    - Document Functional Requirements
    - Document Non-functional Requirements
    - Create Use Cases and User Stories
    - SRS document and project plan
3. Design Phase:
    - User Experience (UX) Design:
      Wireframing
      Prototyping
      Usability Testing
    - User Interface (UI) Design:
        - UI Layouts
        - Color Schemes
        - Typography
        - Iconography
        - Visual Elements
4. Frontend Development:
    - User Registration Module
    - User Profile Management
    - Restaurant Discovery Interface:
    - Location-based Search
    - Filters (Cuisine, Price, Ratings)
    - Restaurant Details
    - Attractions and Points of Interest Interface
    - Travel Booking Functionality
    - Search for Travel Partners Feature
    - Community Interaction
    - User Feeds
    - Real-Time Weather Integration
5. Backend Development:
    - User Data Management

- User Registration and Authentication
- Booking
- API Development for Web Interfaces
- Integration with External Services
- Real-Time Weather APIs

6. Testing and Quality Assurance:

    - Unit Testing
    - Integration Testing
    - User Acceptance Testing
    - Performance Testing
    - Security Testing
    - Bug Tracking and Resolution

7. Deployment:

    - Web Interface Deployment
    - Server Setup and Configuration
    - Database Deployment

8. Documentation and User Guides:

    - User Documentation
    - API Documentation
    - Maintenance and Support Guidelines

9. Maintenance and Support:

    - Ongoing Bug Fixes and Updates
    - User Support
    - Continuous Monitoring and Performance Optimization
    - Regular Data Backups and Security Updates

10. Project Evaluation and Review and Closure:

    - Post-launch Evaluation
    - User Feedback Analysis
    - Iterative Improvements and Updates

ROUGH ESTIMATION OF EFFORT (PERSON-MONTHS):

Considering Data:

- 1 year = 260 working days
- 1 month = 260/12 = 21.66 working days
- 7 full working days for one person = 7/21.66 = 0.323 person-months

Calculating the effort estimations for each task,

1. *Project Kick Off*:
- Effort Estimation = (4 persons × 0.323 person-months/day × 5 days) / 21.66
- Effort Estimation ≈ 0.303 person-months

2. *Research and Gather Information*:
- Effort Estimation = (4 persons × 0.323 person-months/day × 6 days) / 21.66
- Effort Estimation ≈ 0.364 person-months

3. *Information Architecture*:
- Effort Estimation = (4 persons × 0.323 person-months/day × 6 days) / 21.66
- Effort Estimation ≈ 0.364 person-months

4. *Develop the Database*:
- Effort Estimation = (4 persons × 0.323 person-months/day × 6 days) / 21.66
- Effort Estimation ≈ 0.364 person-months

5. **UI/UX Design (Frontend):**
- Effort Estimation = (4 persons × 0.323 person-months/day × 13 days) / 21.66
- Effort Estimation ≈ 1.933 person-months

6. *Backend Development*:
- Effort Estimation = (4 persons × 0.323 person-months/day × 4 days) / 21.66
- Effort Estimation ≈ 0.258 person-months

7. *Integration and Testing*:
- Effort Estimation = (4 persons × 0.323 person-months/day × 6 days) / 21.66
- Effort Estimation ≈ 0.364 person-months

8. *Deployment & Handover*:

- Effort Estimation = (4 persons × 0.323 person-months/day × 2 days) / 21.66
- Effort Estimation ≈ 0.129 person-months

Now, summing up all these effort estimations:

Total Effort Estimation ≈ 0.303 + 0.364 + 0.364 + 0.364 + 1.933 + 0.258 + 0.364 + 0.129 ≈ 4.279 person-months

So, the total estimated effort required for all the tasks is approximately **4.279 person-months**, assuming four people working seven full working days per person.

GANTT CHART :